

What preparing a banking foundation model entails

Start by checking the data. You can have millions of transactions and still not have a foundation-model problem.

Check the data are fit for purpose

Does customer history actually contain useful signal?



Pretrain on customer behaviour

Learn patterns across histories before introducing a specific task.



Specialise for a business question

Adapt the learned representation to a downstream task.



Benchmark against strong alternatives

Test whether the added complexity actually improves the result.

Do not jump straight to a foundation model. For a specific task, a rule-based approach or a gradient boosting model such as XGBoost may perform better.

What makes customer history fit-for-purpose

A dataset can contain long histories and still be a poor fit for sequence learning if a few current event fields already explain most of the target.

Is there enough behavioural history?

Repeated customer activity over time, not isolated events or snapshots.

Months of spending, transfers and account activity, not a handful of transactions.

Are shortcuts dominating the target?

A simple rule or one current event field should not already explain most of the outcome.

One transaction flag should not almost determine the label.

Does sequence context add value?

History should add information beyond the current event or a few simple summaries.

Event order and context should change the answer beyond recent counts.

Having history is not enough. The history must contain information that simpler representations do not already capture.

Evaluating whether customer history is useful for sequence learning

Two datasets with transaction histories were assessed by asking whether customer history had a meaningful role in the prediction.

SynSEPA

Synthetic instant payment histories with fraud labels.

Every non EU payment to a new beneficiary was fraudulent in the sampled data. No payment to an existing beneficiary was fraudulent.

EU cross border payments to a new beneficiary also had a much higher fraud rate.

IBM / Altman

Synthetic credit card transaction histories across users.

User histories contained thousands of events, and almost all users had enough activity for long sequences.

Those histories made it possible to define forward looking questions based on what customers did next, including future category use and future activity tier.

SynSEPA was a poor fit because current transaction rules explained much of the fraud outcome. IBM / Altman was a better fit because its histories supported forward looking questions where past behaviour could matter.

From pretraining to fine tuning

Pretraining learns reusable patterns from customer history. Fine tuning then adapts what was learned to a specific business question.

PRETRAINING

The model first learns recurring patterns from customer histories before any specific task is defined. For example, it can learn that spending, travel and transfers often appear in recognisable behavioural sequences.

FINE TUNING

Future category use

Will this customer use these categories in the next 30 days?

Future activity tier

How active will this customer be in the next 30 days?

The same behavioural representation can be adapted to different business questions without learning customer behaviour from scratch each time.

Comparing different modelling strategies

The comparisons help separate the value of recent behaviour, engineered history features, sequence learning and pretraining.

Recent behaviour only

Persistence baseline

Uses the customer's most recent behaviour as a simple reference point.

Is recent behaviour already enough?

Engineered history features

Gradient-boosted trees

Uses aggregated features derived from past behaviour, such as activity, amounts and category patterns.

Can aggregated history features capture the signal?

Sequence learning without pretraining

Transformer trained from scratch

Learns directly from the ordered customer history using examples from this task only.

Does sequence learning work without pretraining?

Sequence learning with pretraining

Pretrained Transformer

First learns general patterns across customer histories, then fine-tunes on examples from this task.

Does pretraining improve the result?

From pretraining to fine tuning

Pretraining learns reusable patterns from customer history. Fine tuning then adapts what was learned to a specific business question.

PRETRAINING

- Parts of customer history are hidden and the model learns to reconstruct them from the surrounding context.
 - The pretrained Transformer outperformed simple baselines based on the most common or most recent value, suggesting it learned more than simple recency heuristics.
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FINE TUNING

- After fine tuning, pretraining added little over the same sequence model trained from scratch, while engineered history features performed best.
- The foundation model case becomes stronger when representations can be reused across many tasks, labels are scarce, or repeated feature engineering becomes costly.